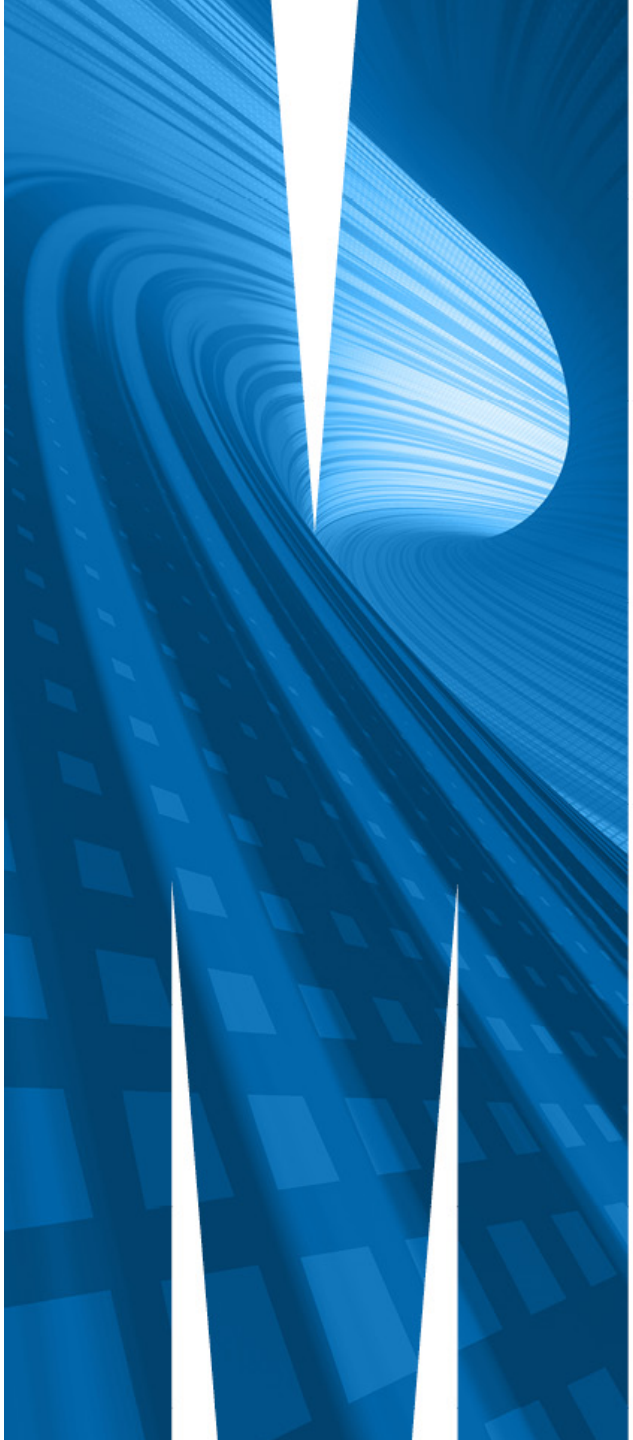
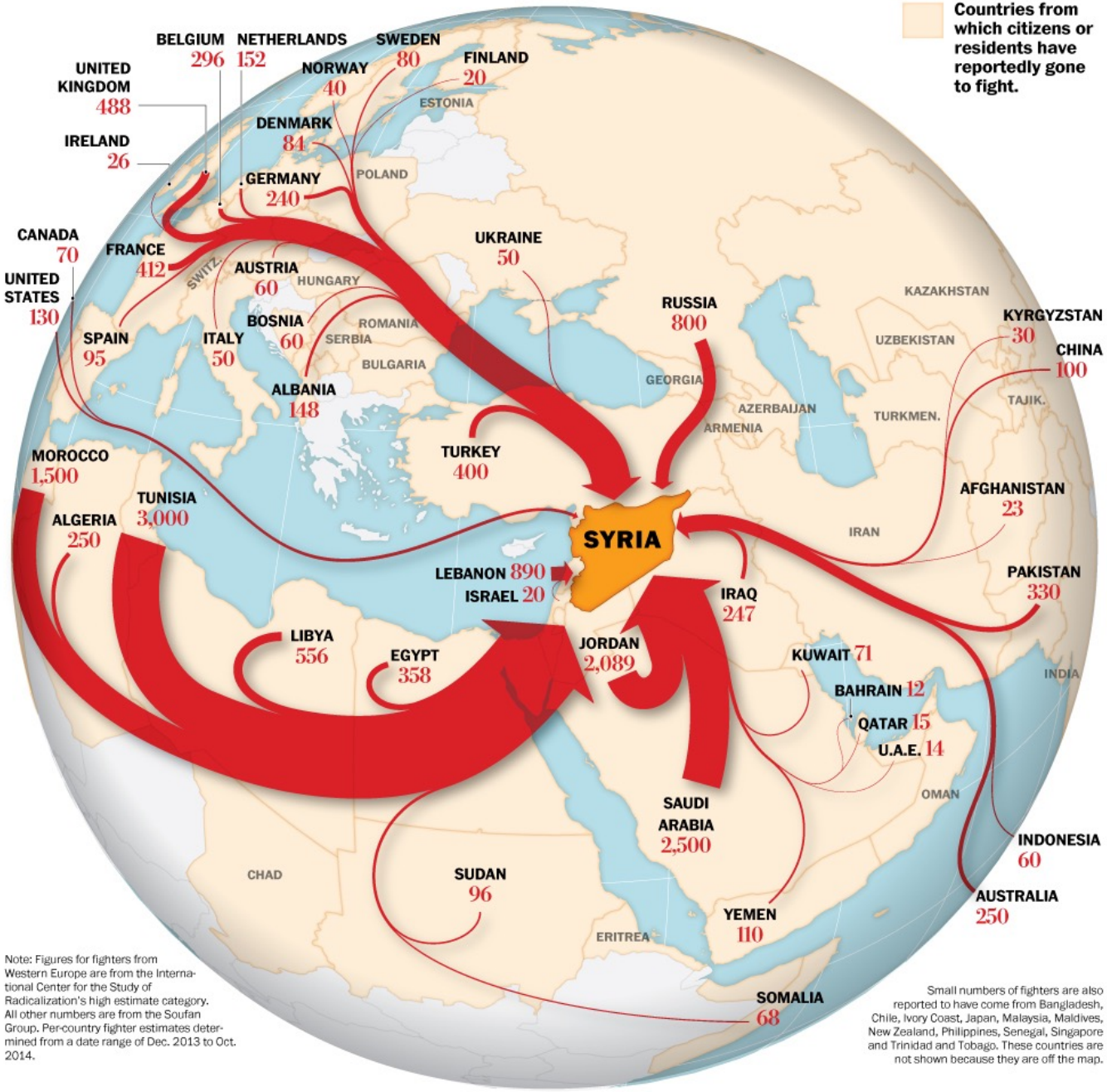
MONASH University

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INFORMATION
TECHNOLOGY

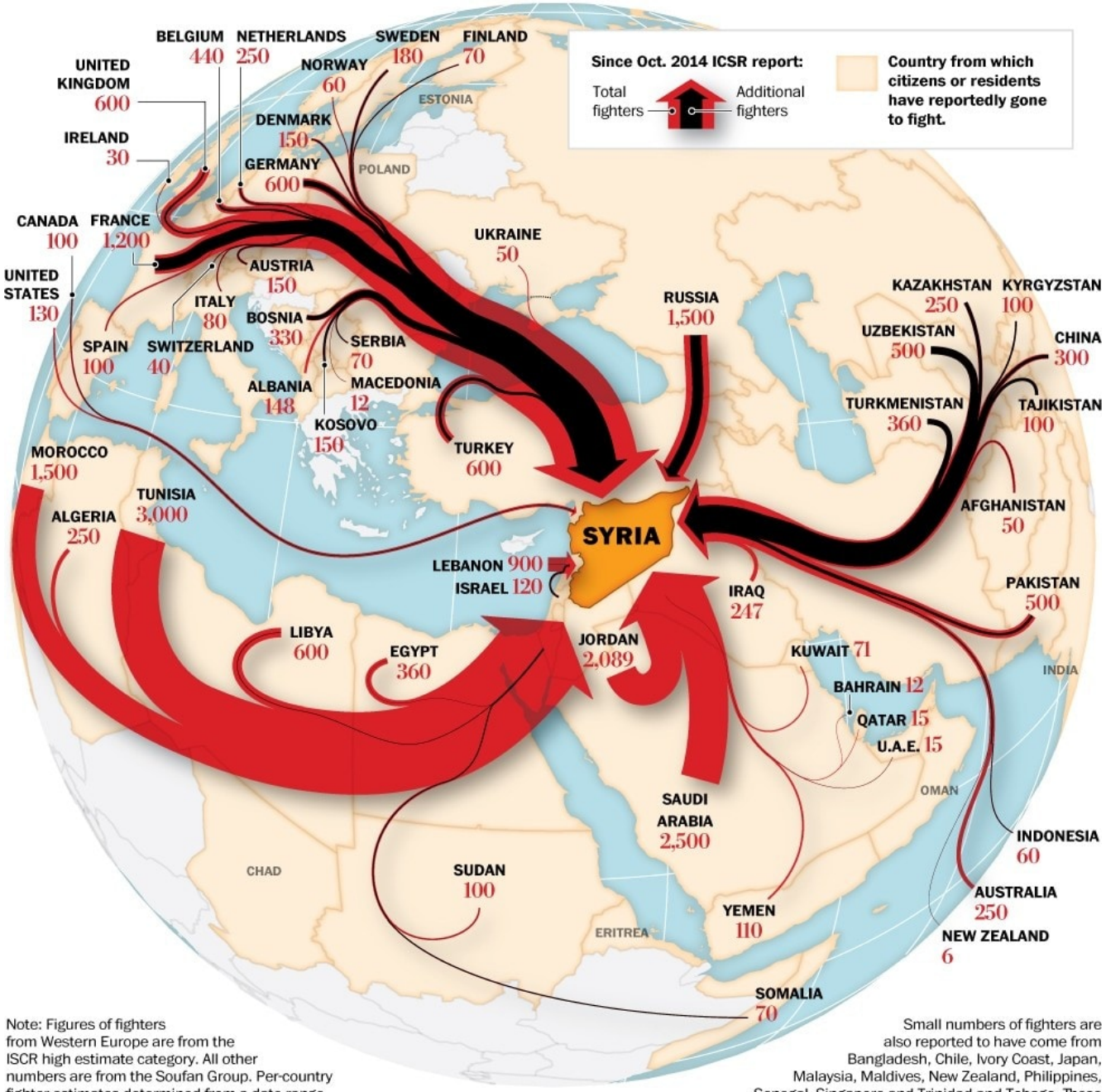


FIT3179 Data Visualisation

Week 6: Discussion of Network and Tree Idioms



The Washington Post: Foreign Fighters Flow to Syria (October 2014)



Note: Figures of fighters from Western Europe are from the ISCR high estimate category. All other numbers are from the Soufan Group. Per-country fighter estimates determined from a date range of December 2013 to the second half of 2014.

Small numbers of fighters are also reported to have come from Bangladesh, Chile, Ivory Coast, Japan, Malaysia, Maldives, New Zealand, Philippines, Senegal, Singapore and Trinidad and Tobago. These countries are not shown because they are off the map.

The Washington Post: Foreign fighters flow to Syria (January 27, 2015)

Questions?

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- Any question about the pre-recorded video lecture?
  1. Networks
     1. Node-link diagram
     2. Chord diagram, alluvial diagram, Sankey diagram
     3. Adjacency matrix
  2. Trees
     1. Dendrogram
     2. Adjacency matrix

Assessments: Project - Data visualisation I

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- Data Visualisation 1.
  - Sample Template, use it for your submission (DV1 template.docx)
  - DV1 Rubric updated - posted on Moodle

    Visualisation idioms and complexity item. Advanced idioms: not simple (from week 5) more than just drag and drop.
  - DV1 Due Sunday Week 6

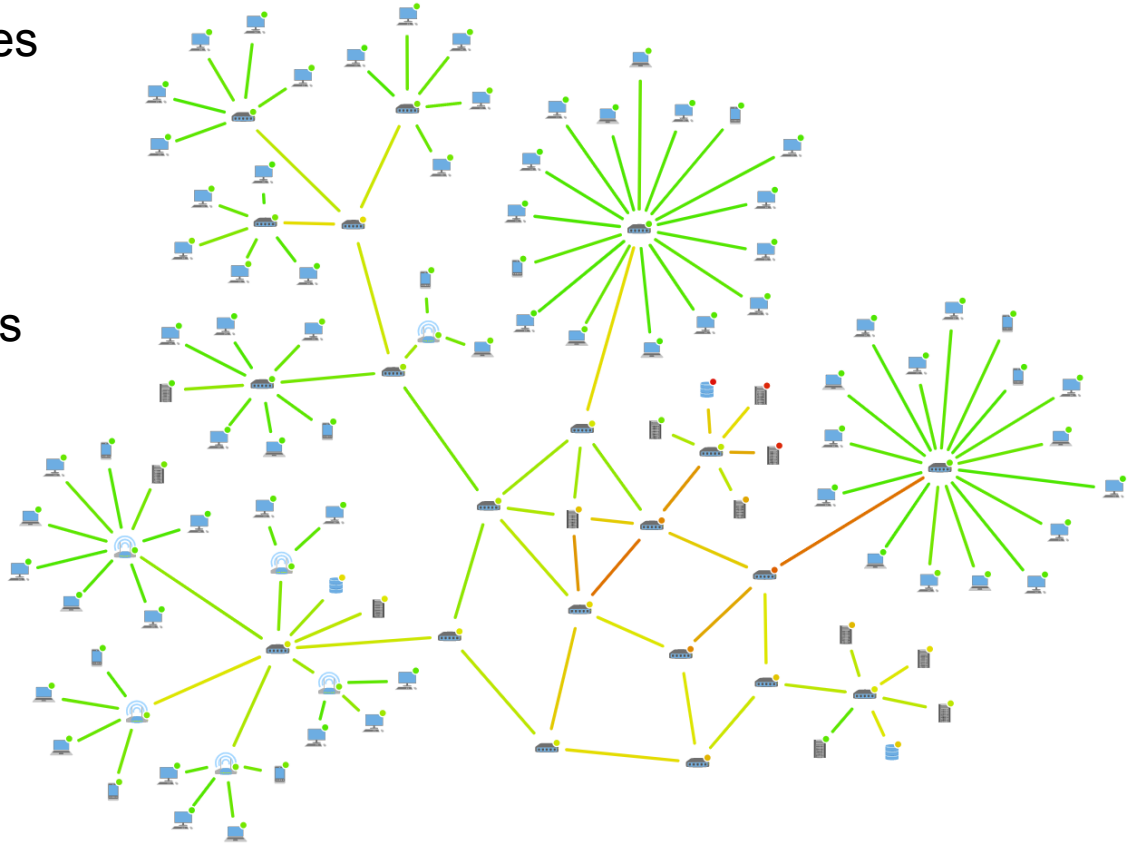
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- Data visualisation I Critique (marked activity 3%)
  - **Purpose**: students to receive feedback on their Data Visualisation 1 assignment from both their peers and tutor, as well as to practise providing constructive feedback.

What is a network or graph?

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- Relationships between entities
- Entities: Nodes or Vertices
- Relationships: Edges or Links
- Graph: network in maths
- Nodes and edges may have attributes (or weights)



Source: https://www.yworks.com/pages/network-monitoring-visualization

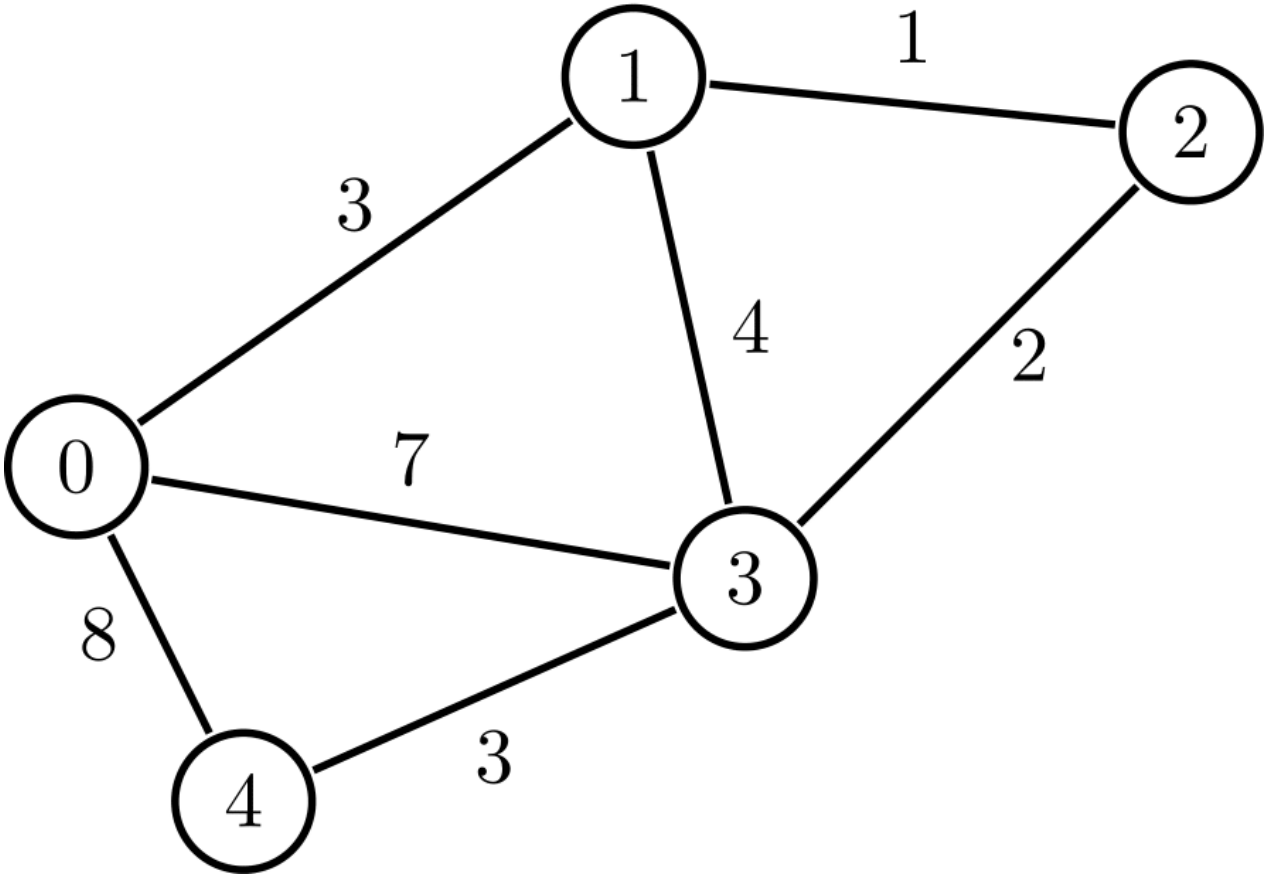
Directed and undirected graphs



Source: https://www.geeksforgeeks.org/what-is-the-difference-between-an-undirected-and-a-directed-graph/

Weighted and unweighted graphs

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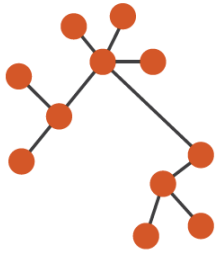


Source: https://hyperskill.org/learn/step/5645

Node–Link Diagrams

Connection Marks

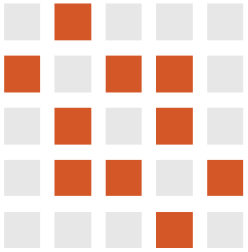
✔ NETWORKS ✔ TREES



Adjacency Matrix

Derived Table

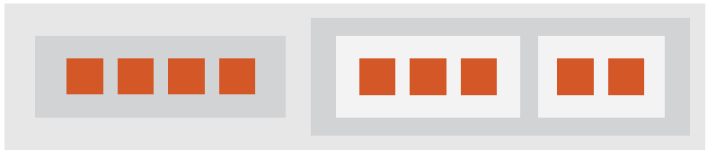
✔ NETWORKS ✔ TREES



Enclosure

Containment Marks

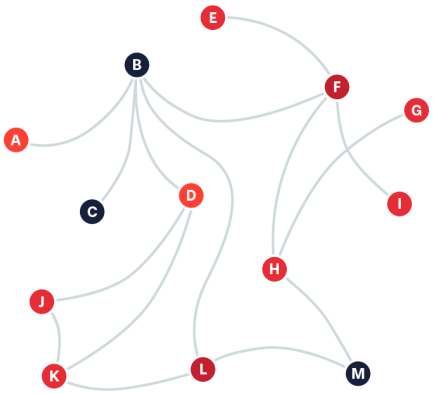
✘ NETWORKS ✔ TREES

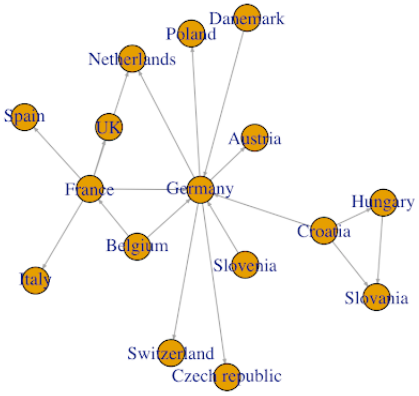


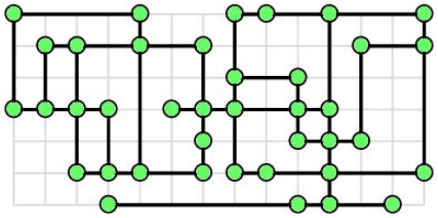
What is a graph drawing (node-link diagram)?

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- Assign each node to a coordinate
- Draw a line between connected nodes to represent edges
- Layout algorithms







Source: https://datavizproject.com/data-type/network-visualisation/

Source: http://www.sthda.com/english/articles/33-social-network-analysis/135-network-visualization-essentials-in-r/

NETWORK DIAGRAM

ALSO KNOWN AS Node–link diagram, graph, hairball graph, social network

C H **R** T S

CONNECTIONS

REPRESENTATION DESCRIPTION

Node–link diagrams display relationships through the connections between data items. The common version of this type of diagram displays items as nodes, represented by point marks, with links or edges (represented by lines) depicting the existence of a connection. These connecting lines will sometimes encode an attribute of direction to indicate the influencer relationship. Replacing point marks with a geometric shape and using attributes of size is a further variation. In some versions a further quantitative weighting is applied to show the relationship strength, maybe through increased line width. Attributes of colour may be used to indicate a quantitative value (e.g. number of connections) and/or some categorical classification.

EXAMPLE Exploring the connections of voting patterns for Democrats and Republicans across all members of the US House of Representatives from 1949 to 2012.

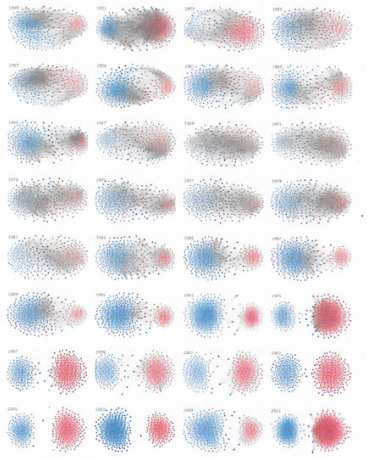


Figure 6.32 The Rise of Partisanship and Super-cooperators in the U.S. House of Representatives, visualisation by Mauro Martino, authored by Clio Andris, David Lee, Marcus J. Hamilton, Mauro Martino, Christian E. Gunning, and John Armistead Selde

VARIATIONS & ALTERNATIVES

There are many derivatives of the node–link diagram, as explained, based on variations in the use of different attributes. 'Hive plots' and 'BioFabric' offer alternative approaches based on replacing nodes with vertices.

PRESENTATION TIPS

INTERACTIVITY: Node–link diagrams are particularly useful when offered with interactive features, enabling the user to interrogate and manipulate the display to facilitate visual exploration. The option to apply filters to reduce the busyness of the display, and maybe enable isolation of individual node connections, can help viewers to focus on specific parts of the network rather than face the whole system at once.

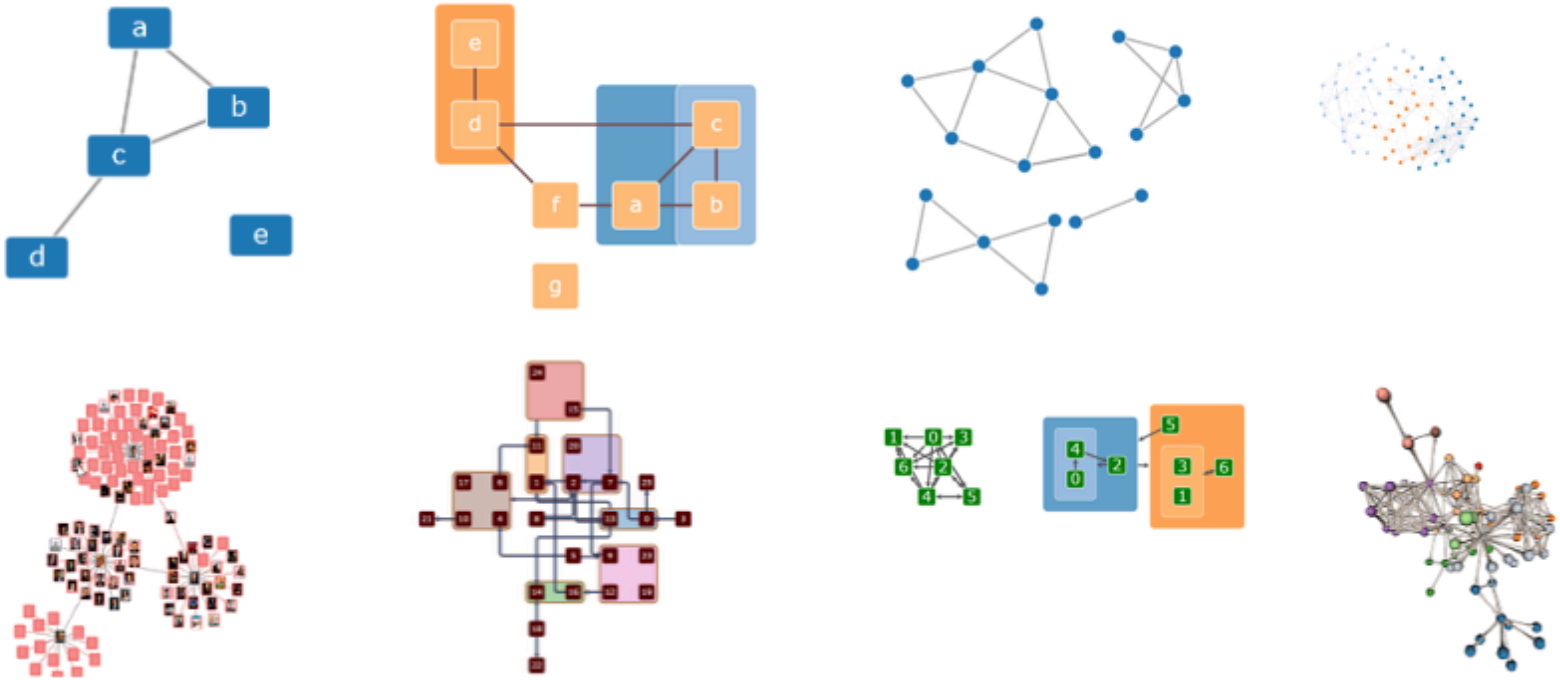
ANNOTATION: The complexity revealed by these diagrams is often a reflection of the underlying subject complexity, so it can be helpful to use annotation to surface key observations about significant clusters or label those items with the most connections.

COLOUR: Aside from the possible categorical colouring of each node, decisions need to be made about the colour of the connecting lines, especially with regard to how prominent these links will be in contrast to the nodes.

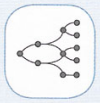
COMPOSITION: Composition decisions will be so varied for any network diagram depending on the complexity and volume of the data and the output constraints around space and consumption format. There are several common algorithmic treatments used to compute custom arrangements to optimise network displays, such as force-directed layouts using the physics of repulsion and springs to amplify relationships. There are also simplifying techniques such as edge bundling to aggregate or summarise multiple similar links.

cola.js

Constraint-Based Layout in the Browser



http://marvl.infotech.monash.edu/webcola



DENDROGRAM

ALSO KNOWN AS Radial tree, layout tree, cluster tree, tree hierarchy

C H R T S

HIERARCHIES

REPRESENTATION DESCRIPTION

A dendrogram is a node–link diagram that displays hierarchical relationships across multiple tiers of categorical dimensions. It displays a hierarchy based on multi-generational 'parent-and-child' relationships. Starting from a singular origin root node (or 'parent') each subsequent set of constituent 'child' nodes, a tier below and represented by points, is connected by lines (curved or straight) to indicate the existence of a relationship. Each constituent node may then have further constituencies represented in the same way, continuing through to the lowest tier of detail.

EXAMPLE Showing a breakdown of the job skills required by workers in the UK job market organised by a hierarchical taxonomy of skill clusters.

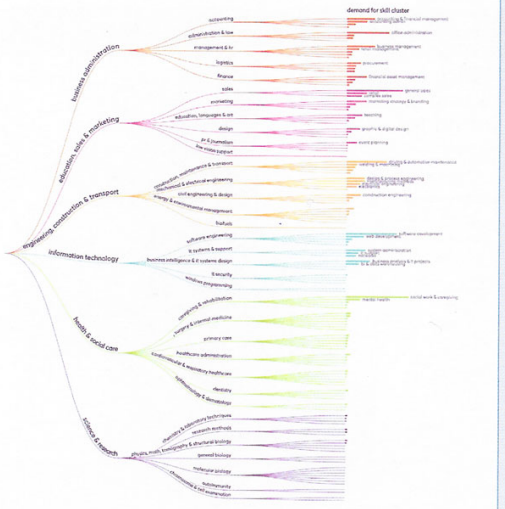


Figure 6.28 Making Sense of Skills: A UK Skills Taxonomy, by Dr Cath Sleeman

VARIATIONS & ALTERNATIVES

Variations of the dendrogram involve incorporating some additional quantitative representation such as using the length or width of connecting lines and, on replacing the point marks for each node, varying the size of node shapes. An alternative approach would be to consider the 'sunburst chart' which would show a part-to-whole relationship across the constituent categories in each hierarchical tier.

PRESENTATION TIPS

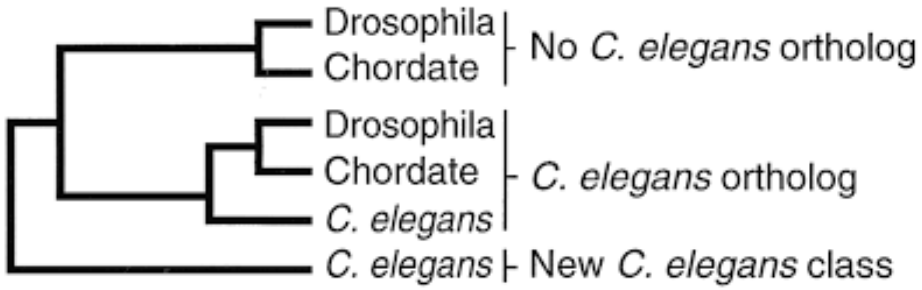
INTERACTIVITY: A useful interactive feature would be to enable filtering or highlighting of branches of interest and selection options for revealing tooltips if labelling is too difficult to accommodate elegantly.

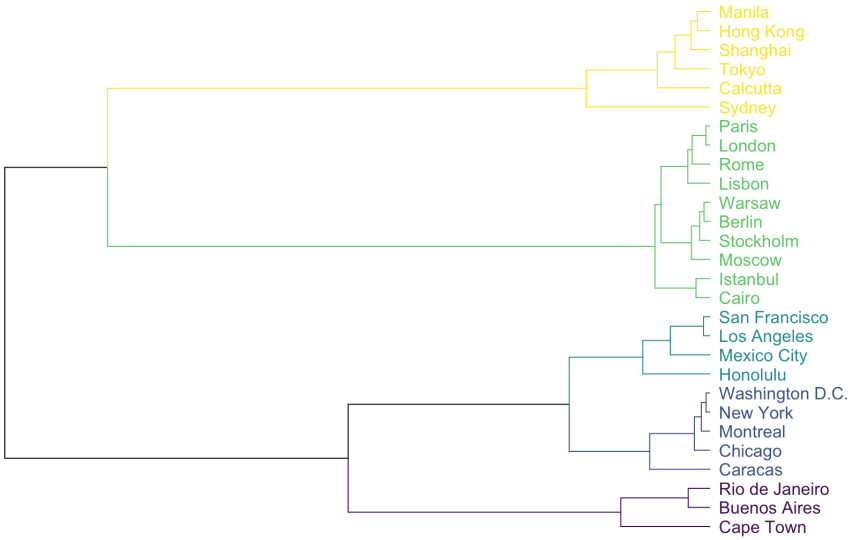
ANNOTATION: If labelling is required, depending on the number of tiers and nodes, the size of the text will need to be carefully considered to ensure readability and minimise the effect of clutter.

COLOUR: Colour would be an optional attribute for accentuating certain nodes or applying further detail of categorisation. The colour of the connecting lines is usually based on a neutral option like black or grey.

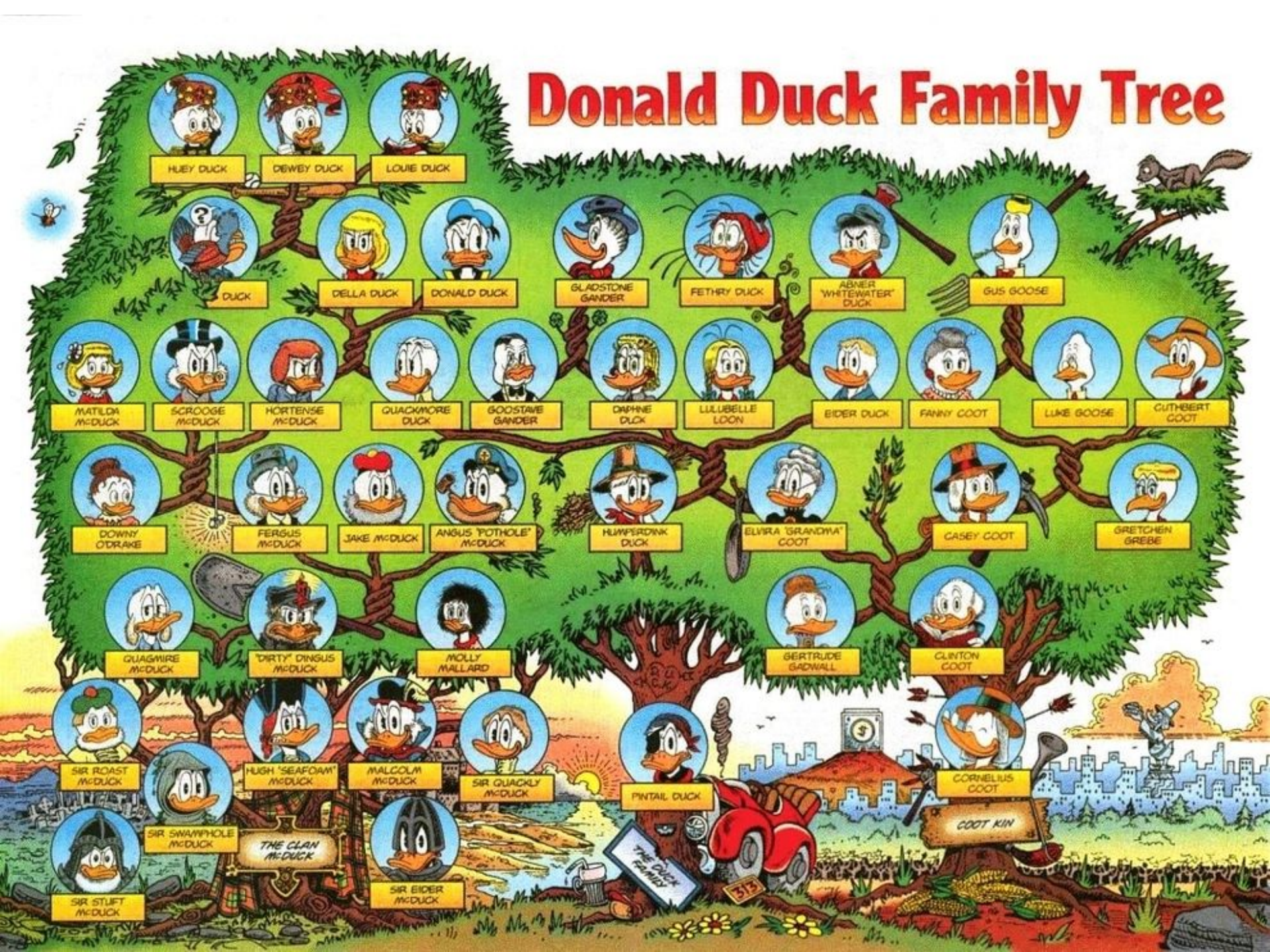
COMPOSITION: The layout can be based on either a linear tree (typically left to right, top to bottom) or radial tree (outwards from the centre) structure. Sometimes, the hierarchical tiers do not necessarily have a parent–child relationship, so their ordering can be legitimately switched around. Therefore, careful decisions are needed about the most logical hierarchical sequencing given the subject matter and enquiry. There is also scope for arranging the sequencing of constituent parts within each tier in a meaningful way.

Dendrogram





https://www.data-to-viz.com/graph/dendrogram.html


Donald Duck Family Tree
HUEY DUCK
DEWEY DUCK
LOUIE DUCK
DUCK
DELLA DUCK
DONALD DUCK
GLADSTONE GANDER
FETHRY DUCK
ABNER "WHITEWATER" DUCK
GUS GOOSE
MATILDA McDUCK
SCROOGE McDUCK
HORTENSE McDUCK
QUACKMORE DUCK
GOOSTAVE GANDER
DAPHNE DUCK
LULUBELLE LOON
EIDER DUCK
FANNY COOT
LUKE GOOSE
CUTHBERT COOT
DOWNY O'DRAKE
FERGUS McDUCK
JAKE McDUCK
ANGUS "POTHOLE" McDUCK
HUMPERDINK DUCK
ELVIRA "GRANDMA" COOT
CASEY COOT
GRETCHEN GREBE
QUAGMIRE McDUCK
"DIRTY" DINGUS McDUCK
MOLLY MALLARD
GERTRUDE GADWALL
CLINTON COOT
SIR ROAST McDUCK
HUGH "SEAFOAM" McDUCK
MALCOLM McDUCK
SIR QUACKLY McDUCK
PINTAIL DUCK
CORNELIUS COOT
SIR SWAMPHOLE McDUCK
THE CLAN McDUCK
COOT KIN
SIR EIDER McDUCK
SIR STUFT McDUCK
THE DUCK FAMILY
313

CHORD DIAGRAM

ALSO KNOWN AS Radial Sankey diagram, radial network, arc diagram

C H **R** T S

CONNECTIONS

REPRESENTATION DESCRIPTION

A chord diagram displays relationships through connections between and within category items. The diagram is formed around a radial display with categories located around the edge: either shown as individual nodes or as arc segments proportionally sized around the circumference to represent a part-to-whole breakdown. Emerging inwards from each origin position are curved lines that join with related categorical destinations around the edge. The connecting lines are proportionally sized according to a quantitative measure and a directional or influencing relationship is often indicated. Attributes of colour hue are commonly used to distinguish different category groupings visually.

EXAMPLE Exploring the connections of migration between and within ten world regions based on estimates across five-year intervals between 1990 and 2010.



Figure 6.34 The Global Flow of People, by Nikola Sander, Guy J. Abel and Ramon Bauer

PRESENTATION TIPS

INTERACTIVITY: Chord diagrams are particularly useful when offered with interactive features, enabling the user to interrogate and manipulate the display to facilitate visual exploration. The option to apply filters to reduce the chaos of the visual and enable isolation of individual or groups of flows helps users to focus on specific relationships of interest. Interactively enabled labelling can also be beneficial as direct labelling is difficulty to incorporate elegantly.

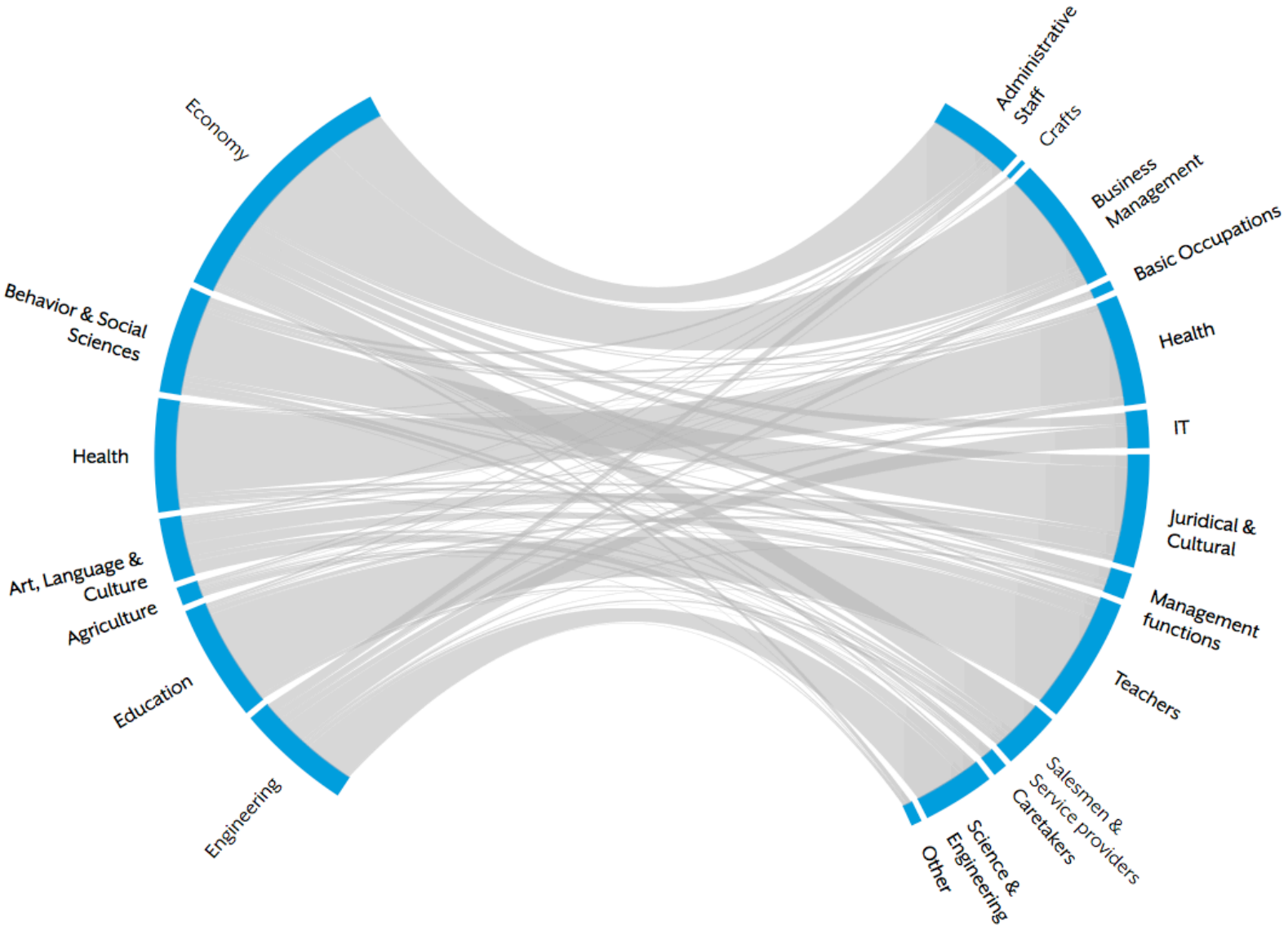
ANNOTATION: Direct labelling will normally be restricted to noteworthy points only. Any colours used must be explained through the inclusion of a legend.

COLOUR: Aside from the categorical colouring of each node, decisions need to be made about the colour of the connecting lines, especially on deciding how prominent these links will be in contrast to the nodes. Sometimes the connections will match the origin or destination colours, or they will combine the two (with a start and end colour blend to match the relationship).

COMPOSITION: The main arrangement decisions concern sorting. Firstly, by deriving as much logical meaning from the categorical values within each stack and, secondly, by deciding on the sorting of the connecting lines in the z-dimension – if many lines are crossing, there is a need to think about which will be on top and which will be below. Showing the direction of connections can be achieved using arrowheads or colour lightness variation. One common, subtle solution is to pull the destination join away from the edge of the destination arc to contrast with connecting lines that emerge directly from an origin.

VARIATIONS & ALTERNATIVES

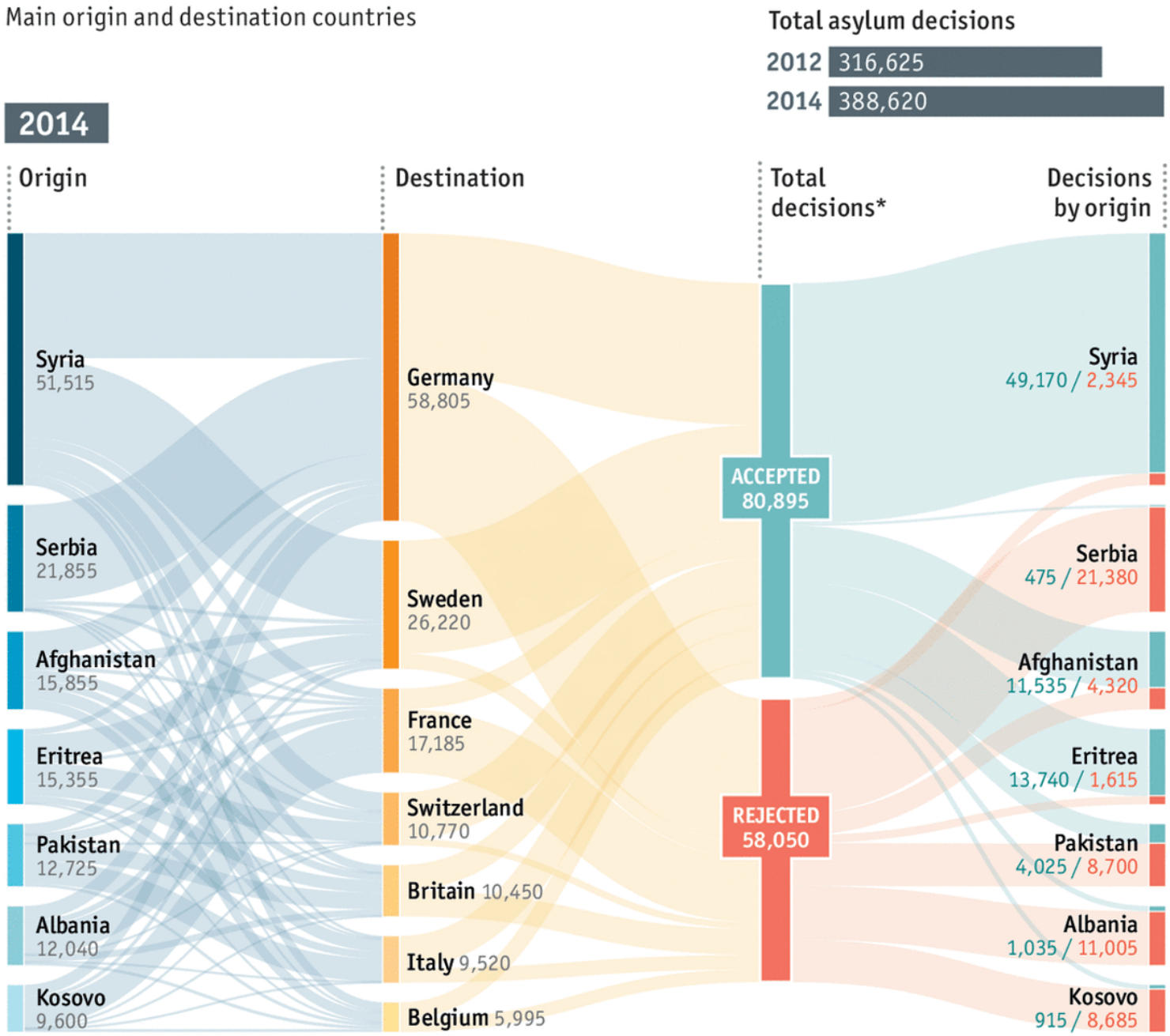
Variations of the chord diagram would be to consider using a single baseline axis, placing all category items along it and forming connections between using semi-circular arcs. Additionally, a 'Sankey diagram' would be relevant if there are distinct origins and destination relationships to reveal.



https://www.visualcinnamon.com/2015/08/stretched-chord.html

European asylum seeker application decisions

Main origin and destination countries



MAGAZINE

See which countries fund the most scientific research

China ranks second only to the U.S.—thanks to a big boost in government spending.

BY MANUEL CANALES AND SEAN MCNAUGHTON

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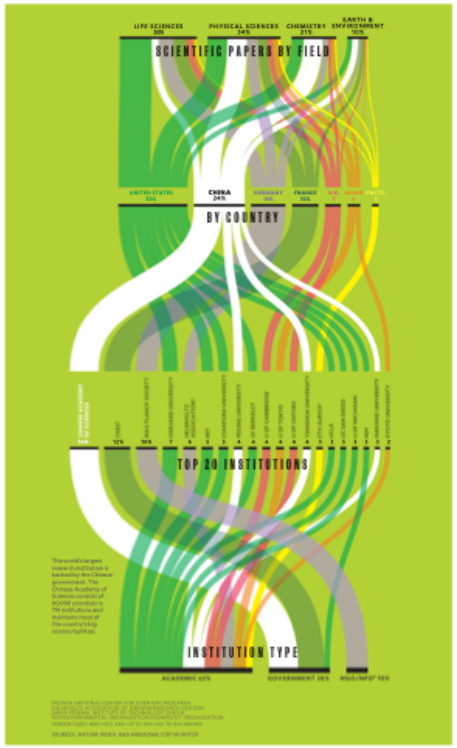


This story appears in the May 2019 issue of National Geographic magazine.

Science is getting a boost in China. Research funding is second only to that of the United States, and the largesse is bearing fruit: The number of papers that Chinese scientists published in major journals rose 17 percent from 2016 to 2018.

Reversing the brain drain

In 2013, nine out of 10 Chinese students were still in the U.S. five years after earning Ph.D.s. In 2017 more than 480,000 scholars in advanced studies abroad returned to China.



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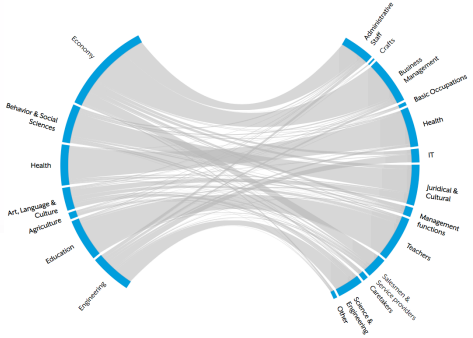
Face of a 7,500-year-old woman reveals Gibraltar's earliest humans

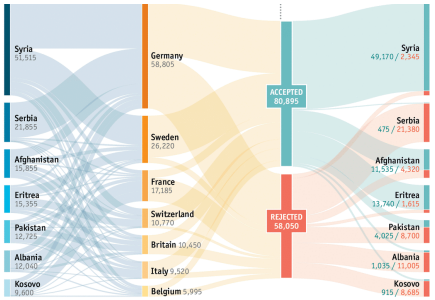


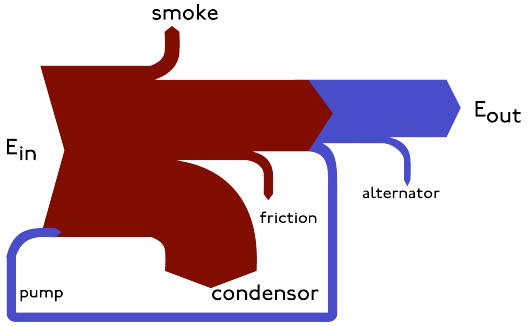
https://www.nationalgeographic.com/magazine/2019/05/data-show-why-china-science-research-is-booming/

Chord Diagram, Alluvial Diagram, Sankey Diagram, Flow Map











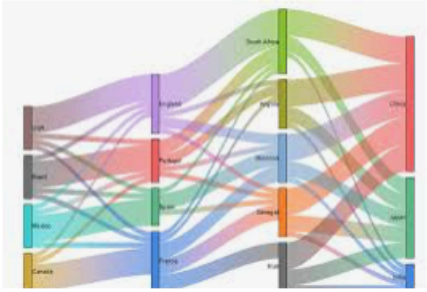
Sankey diagram is often used as umbrella term including chord diagrams and alluvial diagrams.

Google

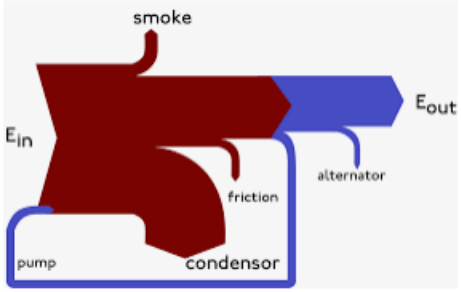
sankey diagram

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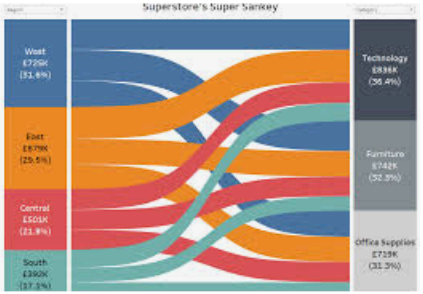
simple | energy | light bulb | tableau | engine | car | efficiency | solar power



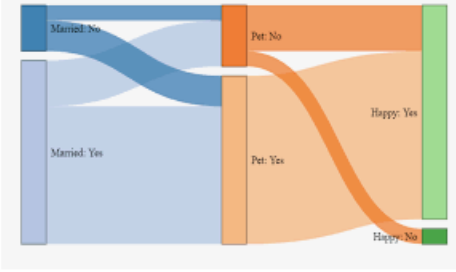
Help Online - Origin Help - Sankey Diagrams
originlab.com



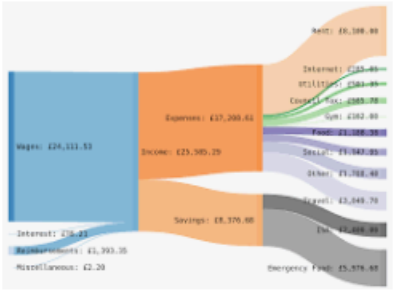
Sankey diagram - Wikipedia
en.wikipedia.org



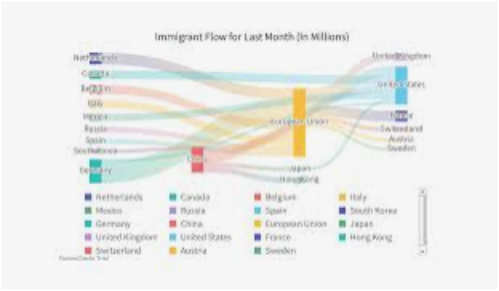
Sankey diagram in Tableau ...
theinformationlab.co.uk



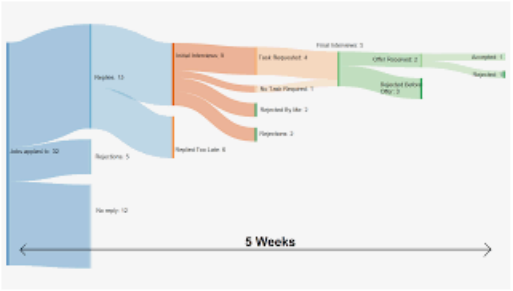
Sankey Diagrams From Tables ...
displayr.com



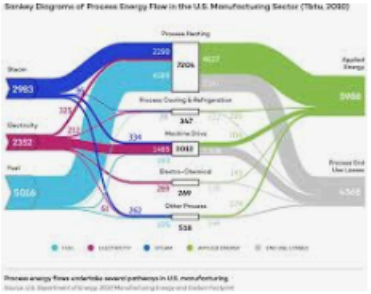
A Sankey diagram showing how I spent ...
reddit.com



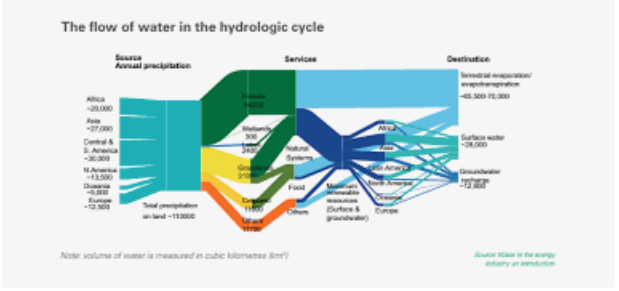
Sankey Diagram | FusionCharts
fusioncharts.com



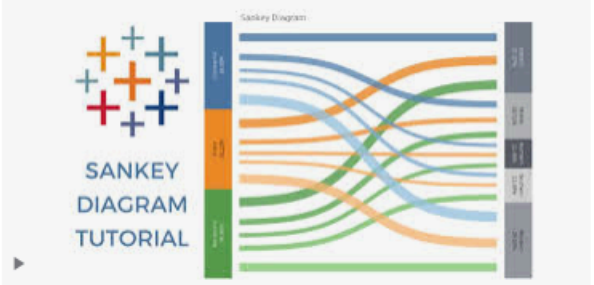
A detailed sankey diagram of my recent ...
pinterest.com



Sankey Diagrams – A Sankey diagram ...
sankey-diagrams.com



Sankey Diagrams ...
towardsdatascience.com



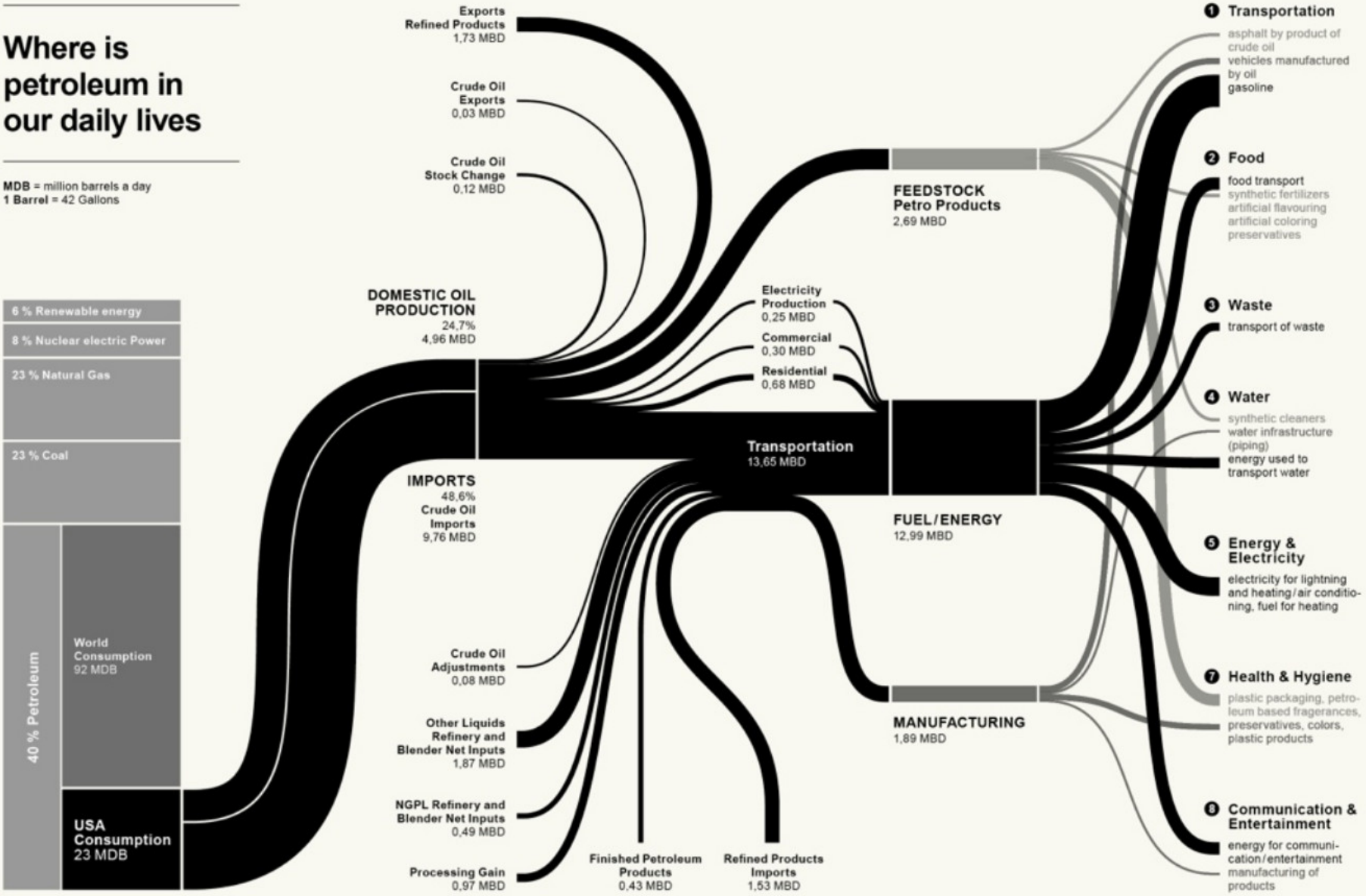
SANKEY DIAGRAM TABLEAU - YouTube
youtube.com



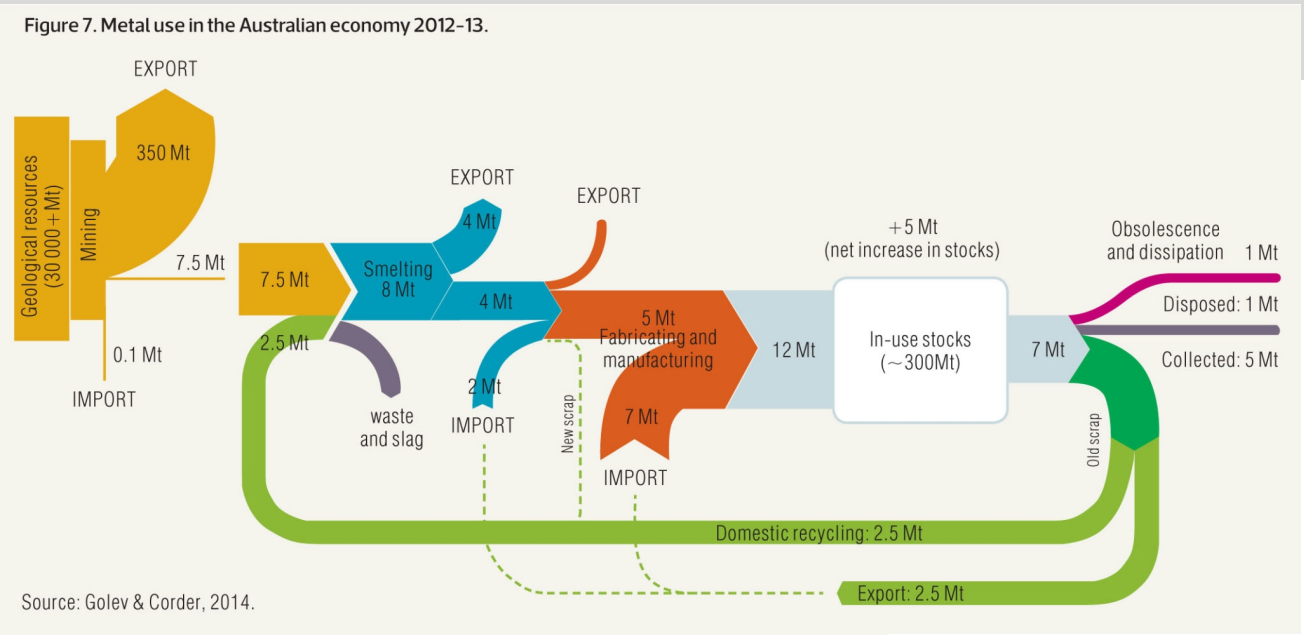
How to Make Sankey Diagram in Tableau ...
towardsdatascience.com

Idiom: Sankey

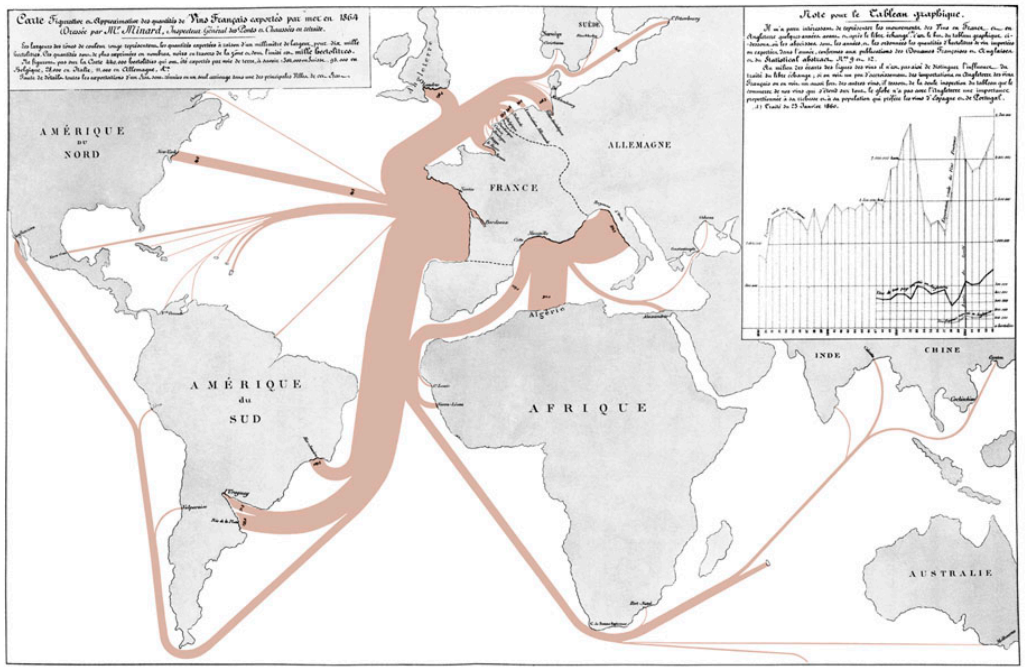
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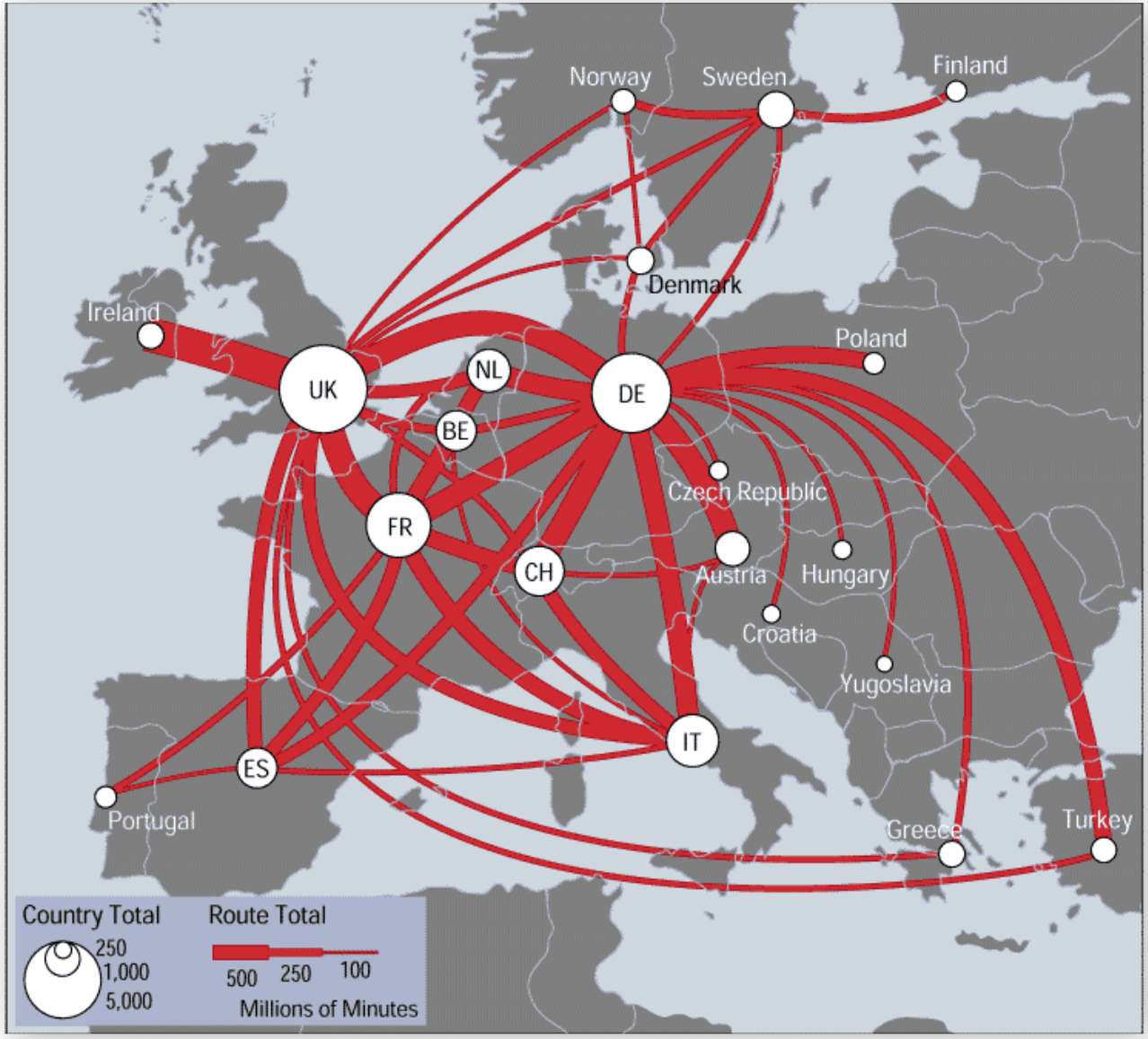
Sankey Diagram vs. Flow Map



Spatial Sankey diagrams are flow maps. Some types of flow map are a spatial Sankey diagram.

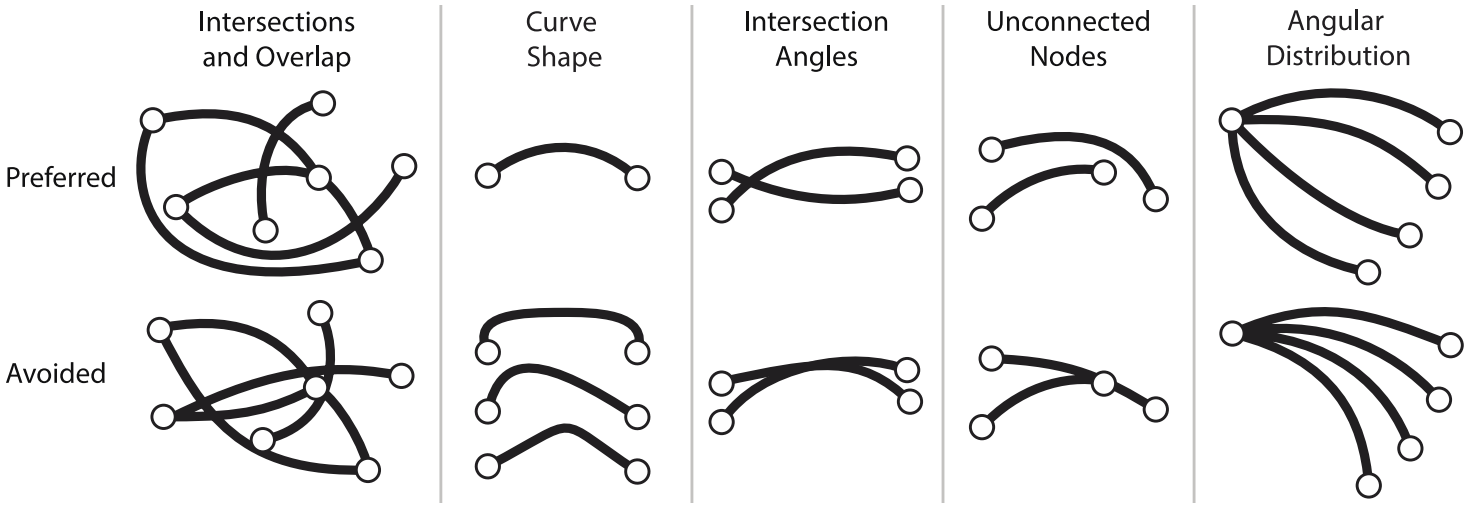


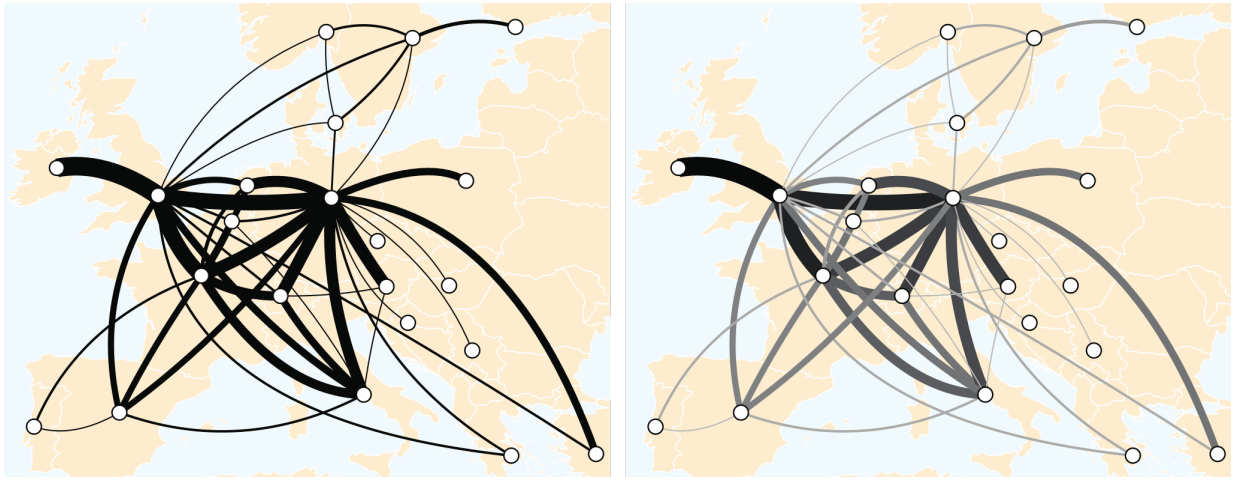
Origin-destination Flow maps



Design Principles for Origin-destination Flow Maps

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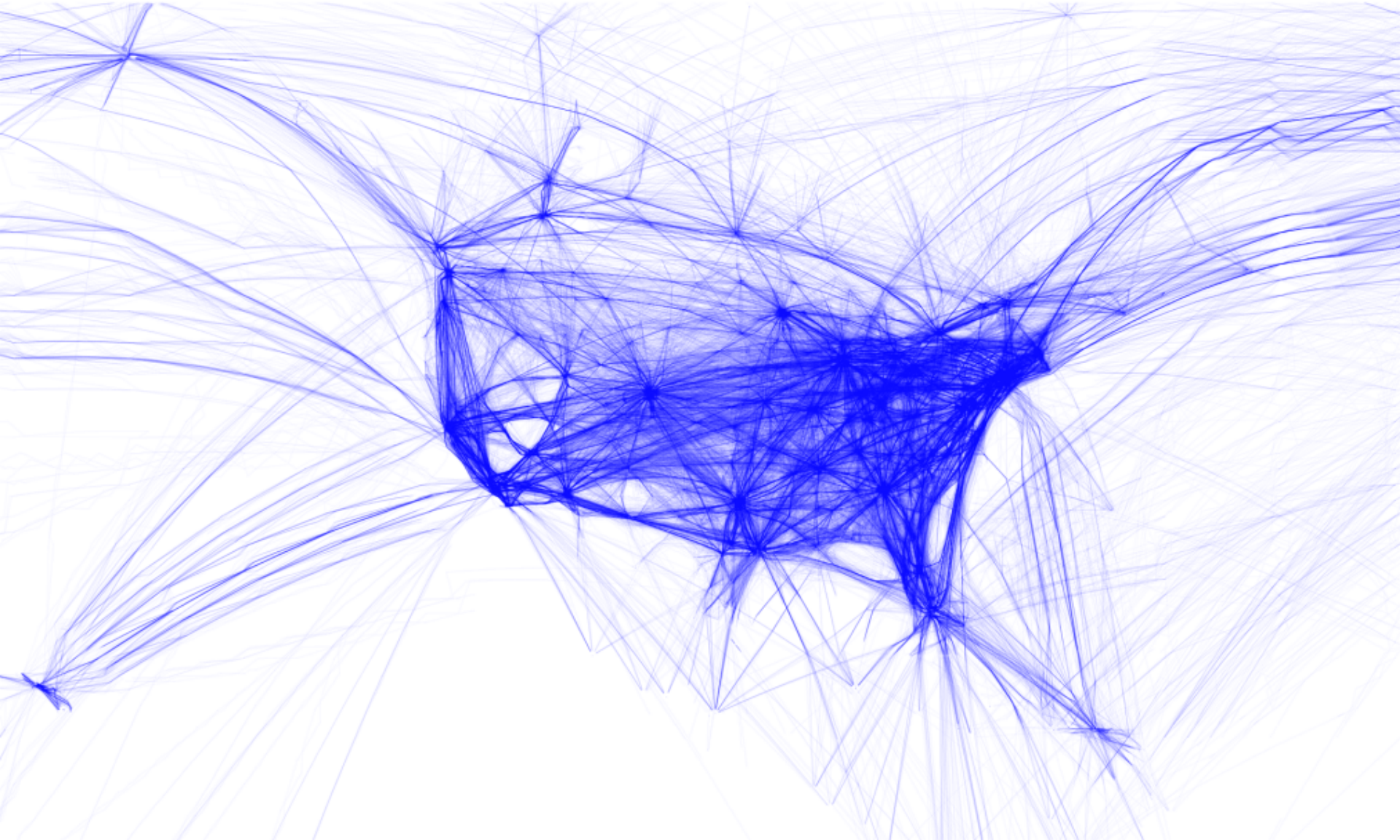




Source: Jenny et al. (2018). Design principles for origin-destination flow maps. Cartography and Geographic Information Science.
http://berniejenny.info/pdf/2017_Jenny_etal_DesignPrinciplesForODFlowMaps.pdf

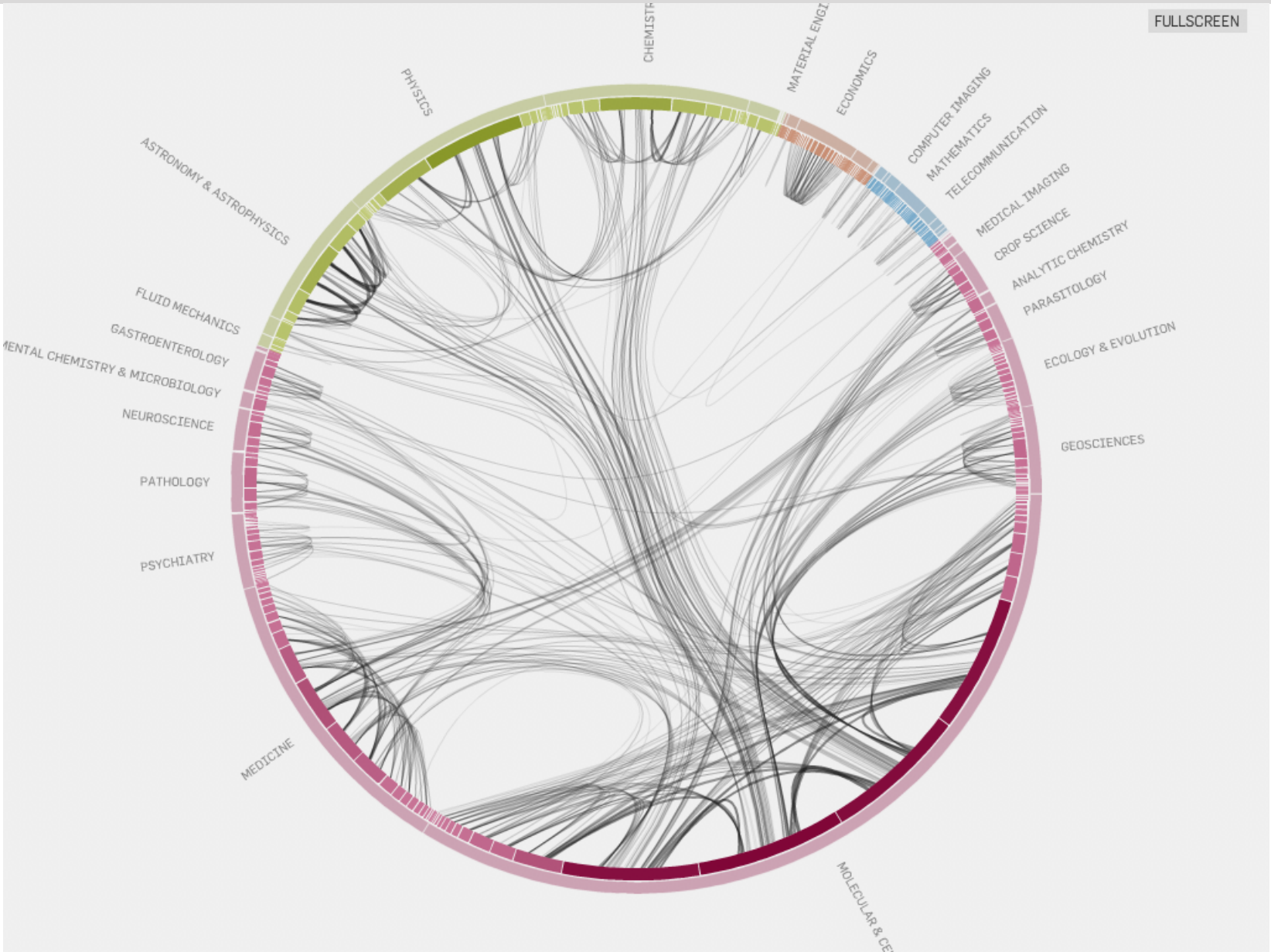
Flow Maps with Path Geometry

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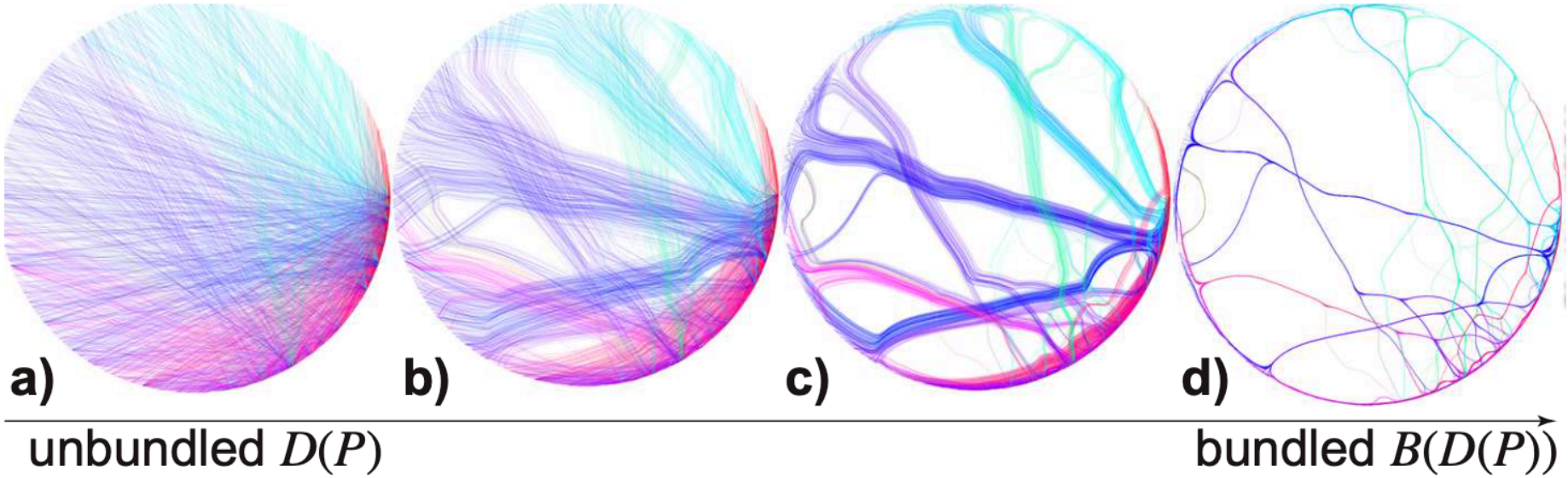
Idiom: Chord Diagram with Bundling

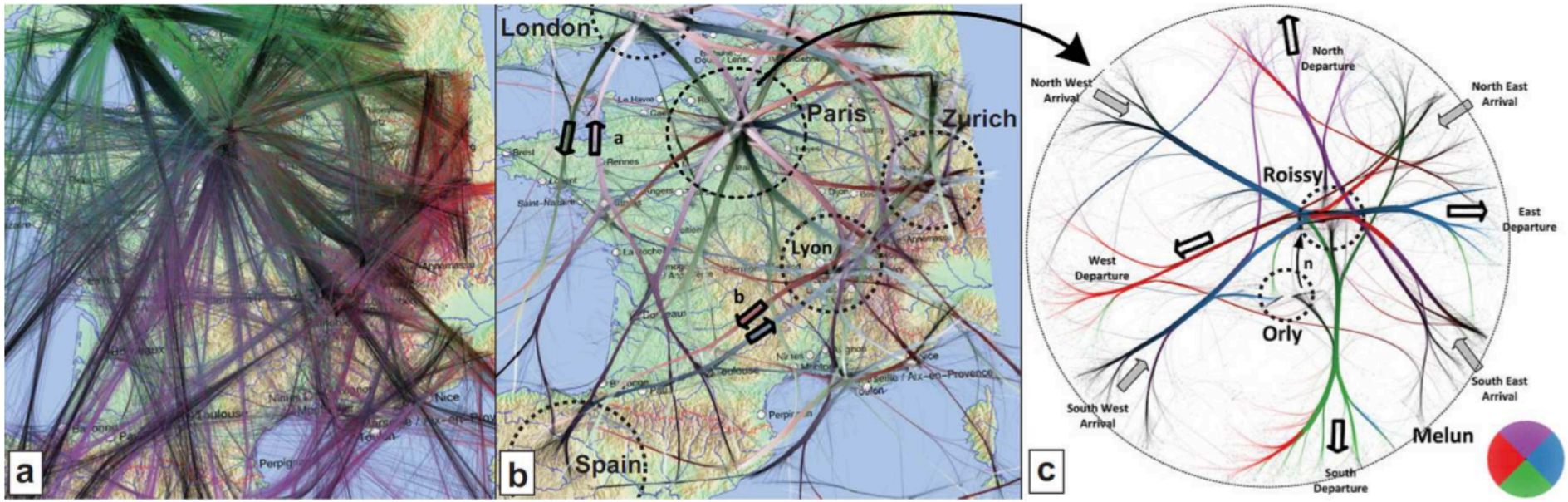
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Edge Bundling

Lhuillier, Hurter & Telea, 2017, State of the Art in Edge and Trail Bundling Techniques





Edge Bundling

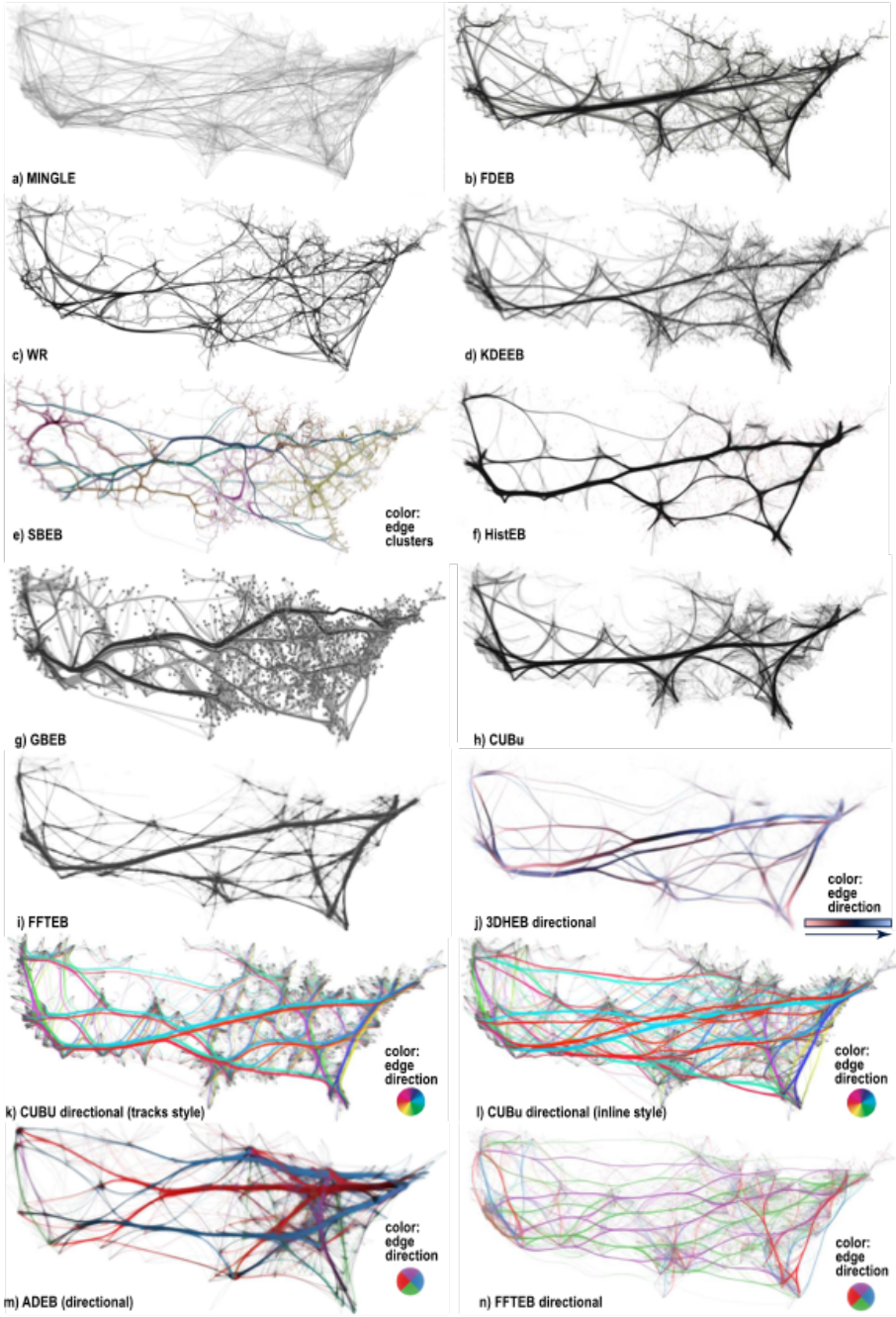
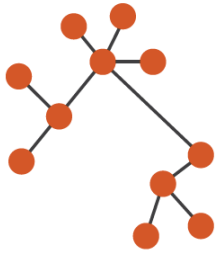


Figure 4: *General undirected (a-h) and directed (i-k) bundling methods,* US migrations *dataset (|N|=1715,|E|=9780). See Sec. 3.2.1.2.*

Node–Link Diagrams

Connection Marks

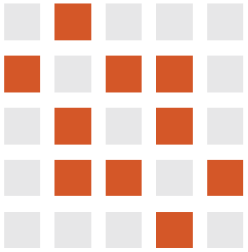
✔ NETWORKS ✔ TREES



Adjacency Matrix

Derived Table

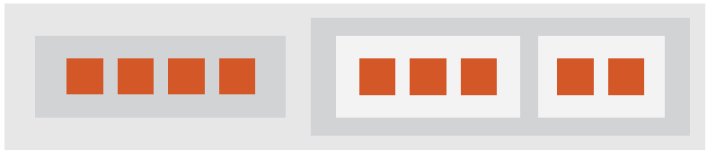
✔ NETWORKS ✔ TREES



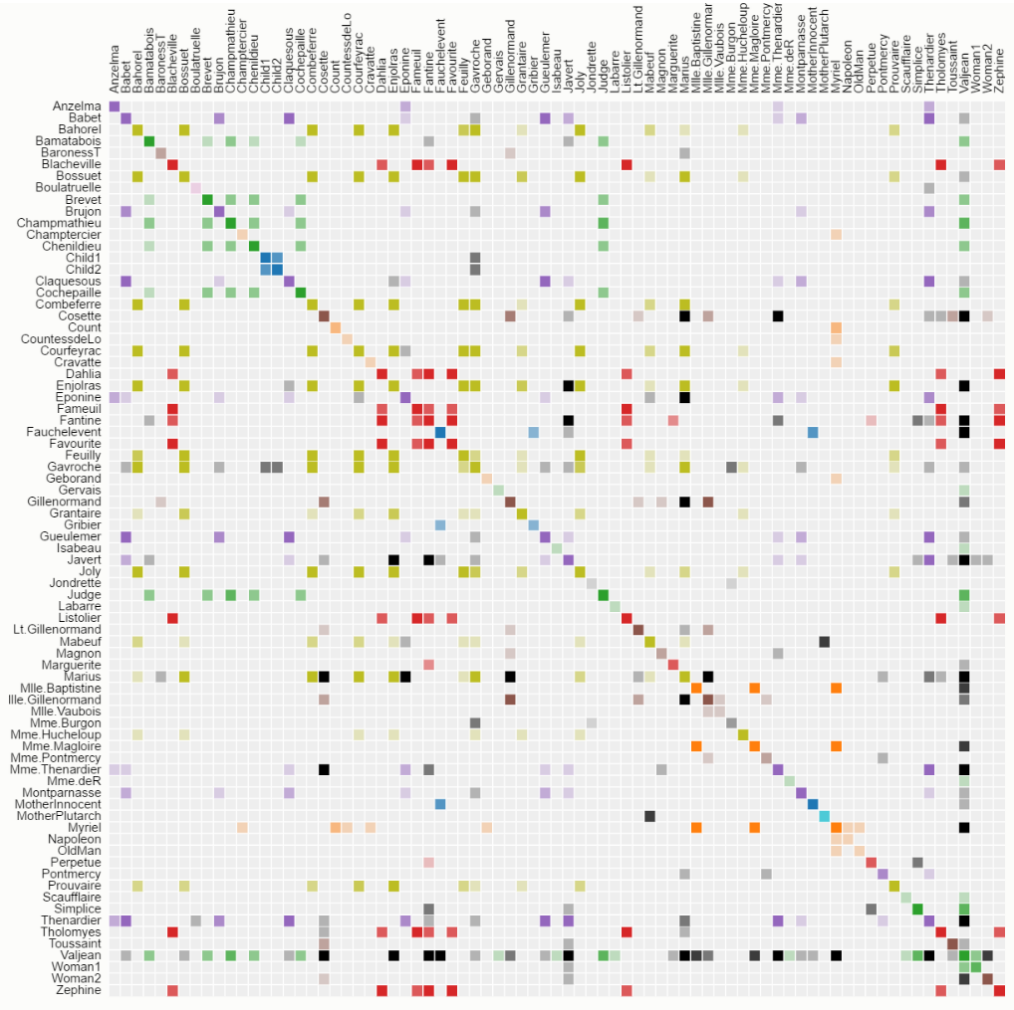
Enclosure

Containment Marks

✘ NETWORKS ✔ TREES



Adjacency Matrix



http://bl.ocks.org/jdfekete/raw/a8828c88060a1883b3f4/

Victor Hugo's Les Misérables

TREEMAP

ALSO KNOWN AS Heat map (wrongly)

C H R T S
HIERARCHIES

REPRESENTATION DESCRIPTION

A treemap is an enclosure diagram providing a hierarchical display that shows how quantitative values for different constituent categorical parts make up a whole. It uses a contained rectangular layout, often termed **squarified**, representing the 100% total. This is divided into proportionally sized shape marks (rectangular tiles) for the quantitative values associated with each categorical part. The organisation of each shape is based on a tiling algorithm to optimise the overall space usage and to cluster related categories into larger rectangle-grouped containers. Attributes of colour are often used to represent further quantitative measures or categorical associations. Treemaps are most commonly used, and of most value, when there are many parts to the whole. The constituent parts must be exclusive and the total representative of a meaningful 'whole', otherwise the chart will be corrupted.

EXAMPLE Comparing the relative value and daily changes of market capital for stocks across the S&P 500 index grouped by sectors and industries.



Figure 6.26 Finviz: Standard and Poor's 500 Index Stocks (www.finviz.com)

PRESENTATION TIPS

INTERACTIVITY: Interactive features that enable selection events to trigger annotated tooltips can be useful, providing direct value labels and details. There may also be scope for modifications to temporal dimensions, changing the sizes and colouring accordingly, or zooming techniques to get a closer view of small constituent parts.

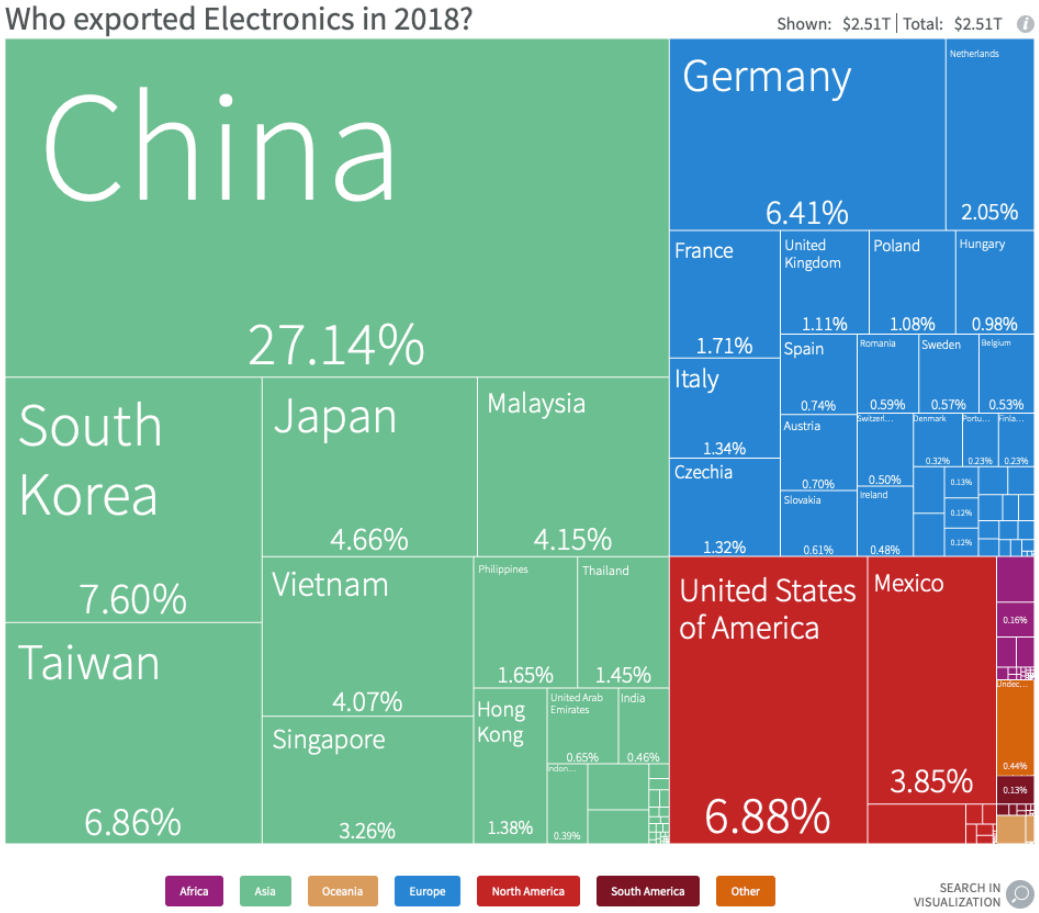
ANNOTATION: Group or container labels can be hard to allocate space to efficiently, so borders are usually applied to indicate the relevant enclosure areas. Effective direct value labelling becomes difficult as the rectangles get smaller, so only the most prominent values may be annotated, especially if interactivity is not available. Any colours used must be explained through the inclusion of a legend.

COMPOSITION: As the tiling algorithm used by any given tool to create a treemap will be focused on optimising the dimensions and arrangement of the rectangular shapes, treemaps may not always be able to facilitate meaningful sorting of high to low values within each enclosure. However, you will generally find larger areas appear in the top left and work outwards towards the smaller constituent parts.

VARIATIONS & ALTERNATIVES

A variation of the treemap sees the overall rectangular layout replaced by a circular one and the tiles represented by polygonal shapes. These are known as 'Voronoi treemaps' as the tiling algorithm is informed by a Voronoi tessellation. The 'circle packing diagram', a variation of the 'bubble chart', similarly shows many parts to a whole but uses non-tessellating circular shapes. The 'Marimekko chart' is similar in appearance to a treemap but, in contrast to the treemap's hierarchical display, presents a breakdown of quantitative percentages and/or absolute values across two categorical dimensions.

Tree Map



http://atlas.cid.harvard.edu/explore/tree_map/import/show/all/0901/2014/